

Production of Hypersonic Sound

In the article titled 'Hypersonic Sound Wherever You Want It' published in Jan. 2008 issue of EFY, the idea is quite revolutionary as far as its implementation is concerned, but the block diagram given in the article does not explain all. Please provide details.

Gautam Anand
Nagpur

The author, T.A. Babu, replies:

Hypersonic sound system is a patent-protected technology of American Technology Corporation. It is not an easy technology to implement into the manufacturing process by simple means. Even some electronics manufacturing giants failed in their attempt to implement because of the complications involved in the hypersonic emitter design. Neither the author nor EFY can provide manufacturing support for such a technology.

If interested, you can contact the manufacturing company directly, which may transfer the technology against its usual terms and conditions. Detailed explanation was not given for the block diagram as the blocks used in this article are standard and have their usual meaning.

PCBs of EFY Projects

I want to assemble the circuits published in EFY and want to know:

1. Where can I find the PCBs of these projects?
2. How can I make the PCBs at home in an easier way?
3. Can I make solder tracks in place of copper tracks on a general-purpose PCB?
4. From where can I get the PCB of the 'Bicycle Indicator' circuit published in July 2008 issue?

Anand Parikh
Ahmedabad

EFY: 1. The PCBs of some of the circuits published in EFY are available from our associates Kits'n'Spares (e-mail: kits@efyindia.com).

2. For making PCBs at home, you may refer to 'Making Home-Brew PCBs' article of EFY April 2006. Or, you can join any short-term course on the

basics of electronics and PCB designing. EFY also conducts such courses at EFY Tech Center, New Delhi.

3. General-purpose PCBs have the copper layer on the solder side. You can make the conductor tracks with wires to extend the connectivity to various points on the solder side.

4. The PCB for the 'Bicycle Indicator' circuit is not available with us.

Cellphone-Operated Land Rover

I have been a subscriber to EFY for the past two years. The circuits published in EFY are quite interesting. I have assembled some of the circuits as hobby projects. These are working well. Now, I have chosen the 'Cellphone-Operated Land Rover' published in July 2008 issue as my academic mini project. I have all the components required for this project. The PCB is ready and I have started soldering the project. However, I couldn't understand how to connect TIP and RING labels (shown at the input of the circuit) using the headset of the cellphone. Please help me.

Sreekanth

Guntur, Andhra Pradesh

☐ I would like to know about the cellphone microcontrollers/processors that may be suitable for the same applications with much more features, in order to miniaturise the circuit and avoid using the external microcontroller. Please suggest.

Maneesh Sharma
Through e-mail

The author, Raghavendra Prasad, replies to Sreekanth:

The input to this circuit is taken from the speaker output of the cellphone. You can connect the headset to the mobile phone and take the connections from the terminals of your headset speaker line, with one wire going to the TIP point and the other wire to ground (RING point). Hope this helps.

Reply to Maneesh Sharma:

I don't know the details of the microcontrollers/processors inside the mobiles. However, I think that your idea of using internal microcontroller

is not feasible. You need an external unit for processing the DTMF signal. Even if you do it inside, how can you send signals for the actuators that are outside your mobile?

Token Number Display for Banks

I am a lecturer in electronics and communication engineering and a regular reader of EFY. I have constructed the 'Bank Token Number Display' project published in April 2008 issue, but without results.

Even after connecting the hardware to the parallel port and running the program, all the 7-segment displays are blank. The hardware is responding well for different inputs without the PC. I have tried to run the parallel port in ECP mode but in vain. Please help.

Subramanyam Avvuru
Through e-mail

The author, Dr Mahesh N. Jivani, replies:

The 'C' program that directly addresses the parallel port will not work under Windows NT, 2000 and XP operating systems. This is because the Windows NT kernel does not allow user-mode programs to have direct access to input/output (I/O) ports. The solution is to write a device driver or use a third-party tool to allow port access.

I have personally used a tool called 'UserPort' program. It is very simple to use for this purpose.

UserPort is a kernel-mode driver that can be used in Windows NT/2000/XP to give user-mode programs access to I/O ports. It is a freeware utility and was included in the EFY-CD of Nov. 2007 issue for use in the 'Parallel Port Interfacing and Programming Aid' construction project.

ERRATA

In the 'Shape Your Career with Microcontroller' article published in Dec. 2008 issue of EFY, please read the starting salary for a fresher in the private sector as Rs 5000 to Rs 30,000 per month, in the third column on page 46.